

Security+ 4.8 · Digital Forensics

Master CyberSec — Official CompTIA Security+ (SY0-701) Comprehensive Study Guide · **Lesson 4.8 — Digital Forensics (9:54)** · [Official YouTube Video ↗](#) · [Professor Messer Study Hub ↗](#)

Executive Lesson Summary: Digital forensics involves the collection, preservation, and analysis of digital evidence. Data collection must follow the order of volatility: volatile data (RAM, cache) first, then disk, then backups. Legal hold ensures evidence is preserved and legal teams are notified. Chain of custody documents who handled evidence, when, and how; without it, evidence may be inadmissible in court.

High-Yield Core Concepts Tested on Exam

- Order of volatility: collect RAM first, then cache, disk, backups
- Legal hold: preserve evidence and notify legal department
- Chain of custody: formal record of every person who handled evidence
- Without chain of custody, evidence may be challenged in court

Exam Tips & Classic Scenarios

- If asked about documenting who handled evidence, the answer is chain of custody.
- Remember: volatile data is lost when the system is powered off.

Typical Exam Scenario: During an investigation, a forensic analyst logs every transfer of a hard drive, including who accessed it and when, to ensure it can be used in court. This is chain of custody.

Beyond the Video: Official CompTIA Exam Concepts, Protocols & Traps (Lesson 4.8 — Digital Forensics)

Why is this advanced coverage required for SY0-701? Video overviews provide foundational familiarity, but CompTIA tests practical engineering judgment and requires deep familiarity with the following protocols, acronyms, and operational nuances in this domain:

- **Principle of Least Privilege:** Users and computational entities must be granted only the minimum permissions strictly required to perform designated tasks.
- **Defense-in-Depth (Layered Security):** Application of multiple security controls (technical, administrative, physical) to prevent single-point perimeter compromise.

Classic CompTIA Traps & Question Framing:

- Avoid rote memorization: CompTIA exam prompts regularly adjust technical constraints requiring you to defend why one valid security control is the *MOST appropriate* or *FIRST step*.
- Scrutinize question qualifiers: 'without additional licensing costs' favors native OS tools; 'granular administrative command accounting' favors TACACS+ over RADIUS; 'non-repudiation' mandates asymmetric private key signing.

Practice Questions with In-Depth Solutions (10 Questions)

Detailed answer key with step-by-step technical rationale and comprehensive distractor breakdown provided below. Attempt answering independently first.

1. When collecting forensic evidence, the 'order of volatility' dictates that you must capture first:

- A)** The hard drive / disk
- B)** RAM (volatile data)
- C)** Tape backups
- D)** Printed logs

2. Record that documents who touched an evidence, when and why (forensics):

- A)** Chain of custody
- B)** Order of volatility
- C)** Log retention
- D)** Non-repudiation

3. Corporate scenario (Digital Forensics): During a security review and compliance audit aligned with the CompTIA Security+ (SY0-701, Section 4.8) exam, an analyst evaluates the following technical requirement: 'Order of volatility: RAM first (lost on shutdown)'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Allow shared access among administrators with generic credentials
- D)** Implement the corresponding solution: Order of volatility: RAM first (lost on shutdown)...

4. Corporate scenario (Digital Forensics): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'Chain of custody: the formal record of every action with the evidence'. What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Chain of custody: the formal record of every action with the evidence...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

5. Corporate scenario (Digital Forensics): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'documenting who handled the evidence' = chain of custody. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'documenting who handled the evidence' = chain of custody.
- D)** Allow shared access among administrators using generic credentials

6. Corporate scenario (Digital Forensics): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'record that maps incident evidence to court' = chain of custody. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Implement the corresponding solution: 'record that maps incident evidence to court' = chain of custody...
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

7. Corporate scenario (Digital Forensics): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'Order of volatility: RAM first (lost on shutdown).' What is the best recommendation or applicable control classification?

- A)** Implement the corresponding solution: Order of volatility: RAM first (lost on shutdown)...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators using generic credentials

8. Corporate scenario (Digital Forensics): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'Chain of custody: the formal record of every action with the evidence'. What is the best applicable recommendation or control classification?

- A)** Implement the corresponding solution: Chain of custody: the formal record of every action with the evidence...
- B)** Temporarily disable audit logs to accelerate system throughput
- C)** Replace preventive technical controls with informal terms of adherence
- D)** Allow shared access among administrators with generic credentials

9. Corporate scenario (Digital Forensics): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'documenting who handled the evidence'=chain of custody'. What is the best recommendation or applicable control classification?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'documenting who handled the evidence'=chain of custody...
- D)** Allow shared access among administrators with generic credentials

10. Corporate scenario (Digital Forensics): During a security review and compliance audit aligned with the CompTIA Security+ exam (SY0-701, Section 4.8), an analyst evaluates the following technical requirement: 'record that maps incident evidence to court'=chain of custody.'. What is the best recommendation or classification of applicable control?

- A)** Temporarily disable audit logs to accelerate system throughput
- B)** Replace preventive technical controls with informal terms of adherence
- C)** Implement the corresponding solution: 'record that maps incident evidence to court'=chain of custody...
- D)** Allow shared access among administrators with generic credentials



COMPREHENSIVE ANSWER KEY & DISTRACTOR BREAKDOWN

1) B — RAM (volatile data).

✓ **Why it's correct:** The correct answer is to capture RAM (volatile data) first. The order of volatility is a fundamental principle of digital forensics, established in RFC 3227 and tested by CompTIA Security+ SY0-701 in the Incident Investigation and Forensics domain. It dictates that evidence must be collected from most volatile to least volatile, because volatile data disappears quickly upon system shutdown or reboot. RAM contains critical artifacts such as running processes, active network connections, encryption keys, malware resident only in memory (fileless), plaintext credentials, and command history. If the system is powered off before RAM capture, this evidence is irrecoverably lost. Therefore, RAM must be acquired before disks, backups, or printed logs, which are less volatile and can be preserved later.

✗ **Why the others are wrong:**

- *The hard drive / disk:* The hard drive (HD) or SSD is non-volatile storage media, whose data persists after system shutdown. Although it is a rich source of evidence (files, logs, registry, system artifacts), it is less volatile than RAM. The order of volatility requires RAM to be captured first, because powering off the machine to preserve the disk would destroy volatile evidence. Therefore, capturing the disk before RAM violates the forensic principle and can lead to loss of crucial data.
- *Tape backups:* Tape backups are copies of data stored offline or on removable media, with very high persistence and low volatility. They are relevant for recovery, but do not contain the volatile state of the system at the time of the incident. In the order of volatility, tapes are among the last items to be collected, as they do not degrade quickly. Prioritizing them over RAM would be a serious error, as volatile data would be lost while handling backup media.
- *Printed logs:* Logs printed on paper are non-volatile physical evidence, which persists indefinitely if stored properly. They may contain useful information, but represent a static and limited snapshot, often already filtered or summarized. In the order of volatility, printed documents are collected last, as they do not undergo spontaneous change. Capturing them before RAM would divert focus from what is truly lost quickly, compromising the integrity of the investigation.

2) A — Chain of custody.

✓ **Why it is correct:** Chain of custody is a chronological documentary record that tracks all handling of digital or physical evidence, from collection to presentation in court or disposal. It records who collected, who transferred, who received, when, where, why, and under what conditions the evidence was maintained. This record is essential to ensure the integrity and admissibility of evidence in legal or disciplinary proceedings, as it demonstrates that the evidence was not altered, contaminated, or substituted. Without an intact chain of custody, evidence may be deemed inadmissible or its probative value questioned.

✗ **Why the other options are wrong:**

- *Order of volatility:* Order of volatility is a forensic principle that guides evidence collection from the most volatile to the least volatile (e.g., RAM, cache, network connections, disk, backups). It defines acquisition priority to prevent loss of data that disappears when the system is powered off. It is not a record of who touched the evidence, when, or why; it is a methodological collection guideline, not a custody tracking document.
- *Log retention:* Log retention refers to policies and practices for storing event logs for a determined period, according to legal, regulatory, or operational requirements. This ensures audit information is available for future investigations. However, log retention does not document the chain of custody of a specific piece of evidence; it only defines how long logs are kept. Chain of custody is a separate and detailed record about evidence handling, not about log storage.

duration.

- *Non-repudiation*: Non-repudiation is a security property that ensures a party cannot deny authorship of an action or the origin of a message, typically implemented via digital signatures and certificates. It is an information security concept related to authenticity and integrity, not a forensic custody record. Although chain of custody may use non-repudiation mechanisms to sign records, the two concepts are distinct: non-repudiation is a cryptographic guarantee, while chain of custody is a documentary process of physical and logical tracking.

3) D — Implement the corresponding solution: Order of volatility: RAM first (lost on shutdown)....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Digital Forensics): Order of volatility: RAM first (lost on shutdown).

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and violates compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

4) A — Implement the corresponding solution: Chain of custody: the formal record of every action with the evidence....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Digital Forensics): Chain of custody: the formal record of every action with the evidence. Chain of custody is a critical forensic process that documents the seizure, custody, control, transfer, analysis, and disposition of evidence, ensuring its integrity and admissibility in legal proceedings.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: This directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and it breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators using generic credentials*: This destroys individual accountability and non-repudiation. It makes it impossible to prove who performed critical changes or malicious commands.

5) C — Implement the corresponding solution: 'documenting who handled the evidence' = chain of custody..

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Digital Forensics): 'documenting who handled the evidence' = chain of custody. Chain of custody is the chronological documentation of the seizure, custody, control, transfer, analysis, and disposition of evidence, ensuring its integrity and admissibility in legal proceedings.

✗ Why the others are wrong:

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting

pillar and forensic traceability. Without logs, the team is blind to incident detection and breaks compliance standards (PCI DSS, ISO 27001).

- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls that rely solely on user conduct and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

6) B — Implement the corresponding solution: 'record that maps incident evidence to court' = chain of custody....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Digital Forensics): 'record that maps incident evidence to court' = chain of custody.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or runtime malware.
- *Allow shared access among administrators with generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

7) A — Implement the corresponding solution: Order of volatility: RAM first (lost on shutdown)....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Digital Forensics): Order of volatility: RAM first (lost on shutdown). This principle dictates that when collecting evidence, the most volatile data (such as RAM, cache, and network connections) should be captured first because it is lost when the system is powered off or rebooted.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput:* Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence:* Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators using generic credentials:* Destroys individual accountability and non-repudiation. Makes it impossible to prove who made critical changes or malicious commands.

8) A — Implement the corresponding solution: Chain of custody: the formal record of every action with the evidence....

✓ **Why it is correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Digital Forensics): Chain of custody: the formal record of every action with the evidence. This is crucial for ensuring the integrity and admissibility of evidence in legal proceedings.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

9) C — Implement the corresponding solution: 'documenting who handled the evidence'=chain of custody....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Digital Forensics): 'documenting who handled the evidence'=chain of custody.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detecting incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.

10) C — Implement the corresponding solution: 'record that maps incident evidence to court'=chain of custody....

✓ **Why it's correct:** According to the official CompTIA Security+ (SY0-701) exam objectives for lesson 4.8 (Digital Forensics): 'record that maps incident evidence to court'=chain of custody.

✗ **Why the others are wrong:**

- *Temporarily disable audit logs to accelerate system throughput*: Directly violates the Accounting pillar and forensic traceability. Without logs, the team is blind to detect incidents and breaks compliance standards (PCI DSS, ISO 27001).
- *Replace preventive technical controls with informal terms of adherence*: Informal terms are administrative controls, rely solely on user conduct, and do not block automated attacks, injections, or malware at runtime.
- *Allow shared access among administrators with generic credentials*: Destroys individual accountability and non-repudiation. Makes it impossible to prove who performed critical changes or malicious commands.